usuario y base de datos

```
mysql> CREATE USER 'userbd1'@'localhost' IDENTIFIED BY 'Password123!';
Query OK, 0 rows affected (0.03 sec)

mysql> CREATE DATABASE firstdb; GRANT ALL PRIVILEGES ON firstdb.* TO 'userbd1'@'localhost'; FLUSH PRIVILEGES;
Query OK, 1 row affected (0.01 sec)

Query OK, 0 rows affected, 1 warning (0.00 sec)

Query OK, 0 rows affected (0.01 sec)

mysql> CREATE TABLE salesRep (employeeNumber INT, surname VARCHAR(40), firstName VARCHAR(20), commission TINYINT);
Query OK, 0 rows affected (0.02 sec)

mysql> DESCRIBE salesRep;
+---------------+------------+------+-----+---------+-------+
| Field         | Type       | Null | Key | Default | Extra |
+---------------+------------+------+-----+---------+-------+
| employeeNumber | int        | YES  |     | NULL    |       |
| surname       | varchar(40) | YES  |     | NULL    |       |
| firstName     | varchar(20) | YES  |     | NULL    |       |
| commission    | tinyint    | YES  |     | NULL    |       |
+---------------+------------+------+-----+---------+-------+
4 rows in set (0.00 sec)
```

insertar y consultar

```
mysql> INSERT INTO salesRep VALUES (1,'Rive','Sol',10
    -> ;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds
yntax to use near '' at line 1
mysql> SELECT * FROM salesRep WHERE surname = 'Gordimer'
    -> ;
Empty set (0.00 sec)

mysql> SELECT * FROM salesRep WHERE surname LIKE 'Ser%'
    -> ;
Empty set (0.00 sec)

mysql> SELECT * FROM salesRep ORDER BY commission DESC LIMIT 1;
Empty set (0.00 sec)
```

funciones y actualización

```
mysql> SELECT MAX(commission), AVG(commission), SUM(commission) FROM salesRep;
+-----------------+-----------------+-----------------+
| MAX(commission) | AVG(commission) | SUM(commission) |
+-----------------+-----------------+-----------------+
|            NULL |            NULL |            NULL |
+-----------------+-----------------+-----------------+
1 row in set (0.00 sec)

mysql> SELECT COUNT(*) AS conteo, COUNT(DISTINCT surname) FROM salesRep;
+--------+-------------------------+
| conteo | COUNT(DISTINCT surname) |
+--------+-------------------------+
|      0 |                       0 |
+--------+-------------------------+
1 row in set (0.00 sec)

mysql> UPDATE salesRep SET commission = 12 WHERE employeeNumber = 1
    -> ;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 0  Changed: 0  Warnings: 0

mysql> ALTER TABLE salesRep ADD dateJoined DATE;
Query OK, 0 rows affected (0.03 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

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Herramientas y backups

```
carlos@carlos-HP-ENVY-x360-2-in-1-Laptop-15-ew1xxx:~$ docker exec mysql-db mysqldump -u root -p4170 firstdb > firstdb_backup.sql
mysqldump: [Warning] Using a password on the command line interface can be insecure.
carlos@carlos-HP-ENVY-x360-2-in-1-Laptop-15-ew1xxx:~$ docker exec mysql-db mysqldump -u root -p4170 --all-databases > full_backup.sql
mysqldump: [Warning] Using a password on the command line interface can be insecure.
carlos@carlos-HP-ENVY-x360-2-in-1-Laptop-15-ew1xxx:~$ docker exec -i mysql-db mysql -u root -p4170 firstdb < firstdb_backup.sql
mysql: [Warning] Using a password on the command line interface can be insecure.
```

Creación de tabla:

```
mysql> CREATE TABLE customer ( id INT PRIMARY KEY AUTO_INCREMENT, firstName VARCHAR(30), surname VARCHAR(40));
Query OK, 0 rows affected (0.02 sec)
```

Inserción de datos:

```
mysql> INSERT INTO customer (id, firstName, surname) VALUES
    -> (1, 'Carlos', 'Martnez'),
    -> (2, 'Ana', 'Lpez'),
    -> (3, 'Luis', 'Ramrez');
Query OK, 3 rows affected (0.00 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

Consulta ordenada:

```
mysql> SELECT * FROM customer
    -> ORDER BY surname ASC;
+----+-----------+---------+
| id | firstName | surname |
+----+-----------+---------+
|  2 | Ana       | Lpez    |
|  1 | Carlos    | Martnez |
|  3 | Luis      | Ramrez  |
+----+-----------+---------+
3 rows in set (0.00 sec)
```

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Conteo:

```
mysql> SELECT COUNT(*) AS total_clientes
    -> FROM customer;
+-----------------+
| total_clientes |
+-----------------+
|               3 |
+-----------------+
1 row in set (0.00 sec)
```

Respaldo:

```
carlos@carlos-HP-ENVY-x360-2-in-1-Laptop-15-ew1xxx:~$ docker exec -i mysql-db mysqldump -u root -p firstdb > firstdb_backup.sql
Enter password: 4170
carlos@carlos-HP-ENVY-x360-2-in-1-Laptop-15-ew1xxx:~$ ls -lh firstdb_backup.sql
cat firstdb_backup.sql | head -20
-rw-rw-r-- 1 carlos carlos 1.3K ago  5 12:26 firstdb_backup.sql
-- MySQL dump 10.13  Distrib 8.0.46, for Linux (x86_64)
--
-- Host: localhost    Database: firstdb
-- ------------------------------------------------------
-- Server version       8.0.46

/*!40101 SET @OLD_CHARACTER_SET_CLIENT=@@CHARACTER_SET_CLIENT */;
/*!40101 SET @OLD_CHARACTER_SET_RESULTS=@@CHARACTER_SET_RESULTS */;
/*!40101 SET @OLD_COLLATION_CONNECTION=@@COLLATION_CONNECTION */;
/*!50503 SET NAMES utf8mb4 */;
/*!40103 SET @OLD_TIME_ZONE=@@TIME_ZONE */;
/*!40103 SET TIME_ZONE='+00:00' */;
/*!40014 SET @OLD_UNIQUE_CHECKS=@@UNIQUE_CHECKS, UNIQUE_CHECKS=0 */;
/*!40014 SET @OLD_FOREIGN_KEY_CHECKS=@@FOREIGN_KEY_CHECKS, FOREIGN_KEY_CHECKS=0 */;
/*!40101 SET @OLD_SQL_MODE=@@SQL_MODE, SQL_MODE='NO_AUTO_VALUE_ON_ZERO' */;
/*!40111 SET @OLD_SQL_NOTES=@@SQL_NOTES, SQL_NOTES=0 */;
/*!40103 SET TIME_ZONE=@OLD_TIME_ZONE */;

/*!40101 SET SQL_MODE=@OLD_SQL_MODE */;
/*!40014 SET FOREIGN_KEY_CHECKS=@OLD_FOREIGN_KEY_CHECKS */;
```

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